GRADE 10

ANALYTICAL GEOMETRY 6
PARALLEL AND PERPENDICULAR LINES

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a) Example: Remember

$m = \frac{\Delta y}{\Delta x}$ $\quad // \text{ lines} : m_1 = m_2$

$m = \frac{y_1 - y_2}{x_1 - x_2}$ $\quad \perp \text{ lines} : m_1 \times m_2 = -1$

// 1. Find the equation of the straight line passing through (2; -3) and parallel to $2y + x = 3$.

$2y = -x + 3$

$y = -\frac{1}{2}x + \frac{3}{2}$

$m = -\frac{1}{2}$ $\quad // \; m_2 = -\frac{1}{2}$

$y = -\frac{1}{2}x + c$

$-3 = -\frac{1}{2}(2) + c$

$-3 = -1 + c$

$-2 = c$

$y = -\frac{1}{2}x - 2$

// 2. Find the equation of the straight line passing through (-2; 2) and parallel to theline through (-2; 1) and (3; 4).

$m = \frac{4-1}{3--2}$

$= \frac{3}{5}$

$y = \frac{3}{5}x + c$

$2 = \frac{3}{5}(-2) + c$

$2 = -\frac{6}{5} + c$

$\frac{16}{5} = c$

$y = \frac{3}{5}x + \frac{16}{5}$

$\perp$ 3. Find the equation of the straight line passing through (2; -3) and perpendicular to $2y + x = 3$.

$2y = -x + 3$

$y = -\frac{1}{2}x + \frac{3}{2}$

$m = -\frac{1}{2}$

$\perp$

$\therefore m_2 = 2$

$y = 2x + c$

$-3 = 2(2) + c$

$-7 = c$

$y = 2x - 7$

$\perp$ 4. Find the equation of the straight line passing through (2; -1) and perpendicular to the line through (-3; 2) and (2; 4).

$m = \frac{4-2}{2--3}$

$= \frac{2}{5}$

$m_2 = -\frac{5}{2}$

$y = -\frac{5}{2}x + c$

$-1 = -\frac{5}{2}(2) + c$

$-1 = -5 + c$

$4 = c$

$y = -\frac{5}{2}x + 4$